

SAUHARDHA KAFLE

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SKILLS

Languages: Rust, TypeScript, Python
Backend: Actix-web, NodeJS, ExpressJS, AIOHTTP, Flask, FastAPI
Frontend: NextJS, ReactJS, TailwindCSS,
Databases & Tools: Redis, MySQL, PostgreSQL, MongoDB, Appwrite
AI/ML: Deep Learning, Natural Language Processing, RAG
Infra: Docker, gRPC, IPC, webRTC, websocket

EDUCATION

Kathmandu University

Dec 2022 – Dec 2026

Bachelor in Computer Engineer

PROJECTS

[nutshell](#) | *TypeScript, Rust, Python, NextJS* | **Personal Project**

- A developer platform to build and deploy web application on cloud.
- Uses **Cloudflare's R2 storage**, **docker** to build the code, **redis** for information flow, **Rust** to handle the request, **NextJS** for frontend and **prisma postgres** for database. Completes overall hosting process within (~60 seconds.)

[Torrex](#) | *Rust* | **Personal Project**

- A bittorrent client implementation from scratch in **rust** programming language.
- Downloads data using torrent file or magnet link that supports **TCP connection**.

[CodeRoom](#) | *NodeJS, MySQL, Docker* | **Team Project**

- A **real-time** collaborative coding platform with multi-user editing, chat, and live code compilation
- Where I handled overall backend, **docker** to compile code which improved to (~50ms), **web socket** for real time messaging in **NodeJs**. Other feature includes **Authentication**, **OTP** code and uses **MySQL** for database.

[Aether](#) | *Python, Rust, WebRTC, Websocket* | **Team Project**

- A browser-based remote desktop sharing platform. A end-to-end system for screen capturing and streaming using **webRTC**.
- Backend coordination server manages session signaling, peer connections, and room state video transfer and reflecting mouse pointer.

[dln-rust](#) | *Rust* | **Personal Project**

- A TUI-based asynchronous download manager leveraging IPC (Unix Domain Socket) for inter-process communication.
- Built with **ratatui** for the terminal interface and **tokio** for async runtime. Features concurrent multi-link downloads, pause/resume functionality, and async task monitoring with **tokio-console**.

[Pneumonia Detection](#) | *TensorFlow, Keras* | **Personal Project**

- Implemented a multi-layer **CNN** in **TensorFlow/Keras** to classify pneumonia from chest X-ray images.
- Includes data preprocessing, training, and evaluation, achieving high recall (~98%) and strong overall accuracy.

EXPERIENCE

Volunteer

- 2023 KURC volunteer
- 2023 KUCC volunteer

Events

- 2024 Event Manager – IdeaPitching
- 2024 Technical Member – ITMeet
- 2025 **Technical Lead** – ITMeet
Led the technical team, We developed AR based marketing, AR based path finder of venue, Website and ensured smoothness of internet throughout the hackathon and other venue.

AI Fellowship – KonnectCraft

2026

Developed an ingestion pipeline for an graphical RAG application that extracts intelligence from the unstructured dataset. Also an webapp for which clients can embed our chatbox in their own product.